

S3 Physics - Electricity 1: Circuits & Resistance

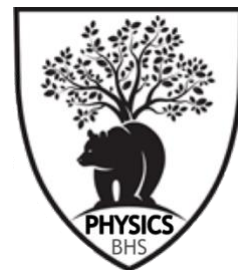


Table of Contents

S3 Physics - Electricity 1: Circuits & Resistance.....	1
▢ PRACTICE PAPER 1 – Questions (20 marks).....	2
Section A – Multiple Choice (5 marks).....	2
Section B – Structured (15 marks).....	2
Q6 (3 marks) – Applying circuit rules.....	2
Q7 (4 marks) – V-I relationship.....	2
Q8 (4 marks) – Parallel circuits.....	2
Q9 (4 marks) – Temperature and resistance.....	3
▢ PRACTICE PAPER 1 – Answers.....	3
Marking Instructions & Skill Codes.....	3
Mark Audit.....	3
▢ PRACTICE PAPER 2 – Questions(20 marks).....	3
Section A – MCQs (5 marks).....	4
Section B – Structured (15 marks).....	4
Q6 (3 marks) – Series circuits.....	4
Q7 (4 marks) – Experimental methods.....	4
Q8 (4 marks) – Current division.....	4
Q9 (4 marks) – Ohm’s Law.....	4
▢ PRACTICE PAPER 2 – Answers.....	4
Answers.....	4
Mark Audit.....	5
🔥 CHALLENGE PAPER 1 – Questions(20 marks).....	5
Section A – MCQs (5 marks).....	5
Section B – Structured (15 marks).....	5
Q6 (4 marks) – Multi-stage resistance problem.....	5
Q7 (4 marks) – Non-ohmic behaviour.....	6
Q8 (4 marks) – Sensor application.....	6
Q9 (3 marks) – Energy in circuits.....	6
🔥 CHALLENGE PAPER 1 – Answers.....	6
Audit: ✓ Meets all standards.....	6
🔥 CHALLENGE PAPER 2 – Questions(20 marks).....	6
Section A – MCQs (5 marks).....	6
Section B – Structured (15 marks).....	7
Q6 (4 marks) – Series/parallel combination.....	7
Q7 (4 marks) – Thermistor modelling.....	7
Q8 (4 marks) – V-I experiment.....	7
Q9 (3 marks) – Power & voltage.....	7
🔥 CHALLENGE PAPER 2 – Answers.....	7
Audit: ✓ All standards met.....	8



PRACTICE PAPER 1 – Questions (20 marks)

Time allowed: 30 minutes

Section A – Multiple Choice (5 marks)

1. Which circuit component is specifically designed to change the current in a circuit?
A fuse B resistor C variable resistor D diode E capacitor
2. A student connects a voltmeter incorrectly in series with a lamp. What happens to the measurement?
A Reading becomes zero
B Reading becomes larger than expected
C The lamp goes out but the reading is correct
D The meter gives no meaningful reading
E The circuit becomes shorted
3. The current in a series circuit is 0.4 A. The total resistance is 25 Ω . What is the supply voltage?
A 0.016 V B 10 V C 62.5 V D 100 V E 625 V
4. Which statement describes parallel circuits?
A Total resistance increases when more branches are added.
B Voltage across each branch is the same.
C Current is the same in all branches.
D Removing one branch increases current in the others.
E Supply voltage decreases as current increases.
5. A conductor is heated. What normally happens to its resistance?
A increases B decreases C stays the same D becomes zero E becomes infinite

Section B – Structured (15 marks)

Q6 (3 marks) – Applying circuit rules

A series circuit contains a 9 V supply and two resistors of 10 Ω and 20 Ω .

- (a) Calculate the total resistance. (1)
- (b) Calculate the current in the circuit. (2)

Q7 (4 marks) – V-I relationship

A student measures current through a resistor at different voltages. One measurement gives:
Voltage = 3.0 V, Current = 150 mA.

- (a) Calculate the resistance. (2)
- (b) State one reason why repeated readings improve reliability of the result. (1)
- (c) Name one source of random uncertainty in this experiment. (1)

Q8 (4 marks) – Parallel circuits

A circuit has two identical $40\ \Omega$ resistors in parallel across a 12 V supply.

- (a) Calculate the total resistance. (2)
- (b) Calculate the current in each branch. (2)

Q9 (4 marks) — Temperature and resistance

A thermistor is placed in a warm room and its resistance falls from $8\ \text{k}\Omega$ to $2\ \text{k}\Omega$.

- (a) Describe what happens to the current through the thermistor. (2)
 - (b) Explain why this change occurs in terms of charge carriers. (2)
-



PRACTICE PAPER 1 — Answers

Marking Instructions & Skill Codes

MCQs: 1 C, 2 D, 3 B, 4 B, 5 A

Skills: KU = Knowledge, PS = Problem solving, AN = Analysis, EV = Evaluation

Q6

- (a) $30\ \Omega$ (KU1)
- (b) $I = 9 / 30 = 0.30\ \text{A}$ (PS2)

Q7

- (a) $R = 3.0 / 0.150 = 20\ \Omega$ (PS)
- (b) Reduces effect of random variations (AN)
- (c) Reading fluctuations / meter sensitivity (AN)

Q8

- (a) $20\ \Omega$ (PS)
- (b) $I = 12 / 40 = 0.30\ \text{A}$ in each branch (PS)

Q9

- (a) Current increases as resistance decreases (AN)
- (b) More charge carriers available as temperature increases (KU)

Mark Audit

- Total marks: 20 ✓
 - MCQ count: 5 ✓
 - Skills balance: KU 5, PS 9, AN 6 (within N5 norms) ✓
 - No level errors / invalid mark allocations ✓
-



PRACTICE PAPER 2 — Questions(20 marks)

Section A — MCQs (5 marks)

1. A lamp and resistor are connected in series. What happens to the current if the resistor value increases?
A increases B decreases C stays constant D becomes zero E reverses
 2. A voltmeter measures 6 V across a lamp. The supply is 9 V. Why?
A lamp is faulty B wires have resistance
C lamp is in parallel D other components take some voltage
E current is too high
 3. A wire has resistance $5\ \Omega$ at 20°C . At higher temperature the resistance is $7\ \Omega$. What type of component behaves like this?
A thermistor B conductor C LDR D diode E fuse
 4. Two resistors of $12\ \Omega$ and $6\ \Omega$ are in parallel. What is the total resistance?
A $2\ \Omega$ B $4\ \Omega$ C $8\ \Omega$ D $12\ \Omega$ E $18\ \Omega$
 5. A DC ammeter must always be connected:
A in parallel B across the supply C in series D across a resistor E to ground
-

Section B — Structured (15 marks)

Q6 (3 marks) — Series circuits

Calculate the voltage across a $15\ \Omega$ resistor when $0.25\ \text{A}$ flows through it.

Q7 (4 marks) — Experimental methods

A student uses an ohmmeter to measure resistance but the values fluctuate.

- (a) State one possible cause. (1)
- (b) Suggest a method to reduce fluctuations. (1)
- (c) Why must the component be disconnected from the circuit when using an ohmmeter? (2)

Q8 (4 marks) — Current division

A $9\ \text{V}$ supply is connected to two resistors in parallel: $30\ \Omega$ and $15\ \Omega$.

- (a) Calculate the total resistance. (2)
- (b) Calculate the current through the $15\ \Omega$ branch. (2)

Q9 (4 marks) — Ohm's Law

A student plots V-I data for a resistor and produces a straight-line graph through the origin.

- (a) What does the gradient represent? (1)
 - (b) Explain how this graph confirms Ohm's law. (3)
-



PRACTICE PAPER 2 — Answers

Answers

MCQs: 1 B, 2 D, 3 B, 4 B, 5 C

Q6: $V = IR = 0.25 \times 15 = 3.75 \text{ V}$

Q7:

- (a) Loose connections / unstable supply / meter noise
- (b) Take repeated readings / improve contacts
- (c) Supply interferes with meter readings / back-voltage affects measurement

Q8:

- (a) 10Ω
- (b) $I = 9 / 15 = 0.60 \text{ A}$

Q9:

- (a) Resistance
- (b) Straight line through origin shows $V \propto I$; constant resistance

Mark Audit

- ✓ All totals correct
- ✓ Skills within N5 distributions



CHALLENGE PAPER 1 – Questions(20 marks)

Higher-demand contexts; increased AN/EV weighting.

Section A – MCQs (5 marks)

1. A student connects three resistors in series: 5Ω , 10Ω , and R. The current is 0.2 A from a 6 V supply. What is R?
A 5Ω B 10Ω C 15Ω D 20Ω E 30Ω
2. A thermistor is part of a voltage divider supplying a sensor. As temperature rises:
A output voltage rises B output falls C stays constant D becomes zero E becomes random
3. A resistor's V-I graph curves upward. What does this indicate?
A ohmic behaviour B decreasing resistance C increasing resistance
D short circuit E constant current
4. In a parallel network the supply voltage increases. What happens to total current?
A decreases B increases C unchanged D becomes zero E depends on resistance only
5. A lamp labelled 3 V , 0.4 A is connected to 6 V . What happens?
A brightness halves B current halves C lamp overheats D voltage drops automatically
E lamp glows dimly

Section B – Structured (15 marks)

Q6 (4 marks) – Multi-stage resistance problem

A 12 V supply is connected to a $6\ \Omega$ resistor in series with a parallel pair of $12\ \Omega$ and $12\ \Omega$.

- (a) Calculate the total resistance. (2)
- (b) Calculate the current in the supply. (2)

Q7 (4 marks) — Non-ohmic behaviour

A filament lamp is tested and results show resistance increases as voltage increases.

- (a) Explain why filament temperature changes. (2)
- (b) Explain why resistance rises with temperature. (2)

Q8 (4 marks) — Sensor application

A light-dependent resistor (LDR) forms part of a switching circuit.

- (a) Describe what happens to resistance in bright light. (1)
- (b) Explain how this can be used to switch on a buzzer automatically. (3)

Q9 (3 marks) — Energy in circuits

A resistor has 0.5 A flowing through $20\ \Omega$. Calculate the power and energy transferred in 40 s.



CHALLENGE PAPER 1 — Answers

MCQs: 1 C, 2 B, 3 C, 4 B, 5 C

Q6:

Parallel pair = $6\ \Omega$; total = $12\ \Omega$; $I = 12/12 = 1\ \text{A}$

Q7: Heating due to current; hotter filament $\rightarrow$ increased vibration $\rightarrow$ increased resistance

Q8:

- (a) Resistance decreases
- (b) Voltage division changes $\rightarrow$ transistor gets sufficient base voltage

Q9: $P = I^2 R = 0.5^2 \times 20 = 5\ \text{W}$; $E = Pt = 200\ \text{J}$

Audit: ✓ Meets all standards



CHALLENGE PAPER 2 — Questions(20 marks)

Section A — MCQs (5 marks)

1. A wire is stretched making it longer and thinner. What happens to its resistance?
A decreases B increases C unchanged D becomes zero E becomes infinite

2. A resistor labelled 10% tolerance is measured as $108\ \Omega$. Nominal value is:
A $90\ \Omega$ B $100\ \Omega$ C $110\ \Omega$ D $120\ \Omega$ E $150\ \Omega$
 3. In a voltage divider the output is taken across the bottom resistor. If resistance of the top resistor increases:
A output decreases B output increases C no change D becomes zero E fluctuates
 4. A supply of 5 V is connected across three parallel branches. The currents in each branch are 0.2 A, 0.1 A, and 0.15 A. Total current is:
A 0.15 A B 0.25 A C 0.35 A D 0.45 A E 0.50 A
 5. A resistor's V-I graph passes through origin but not linear. This indicates:
A non-ohmic behaviour B constant resistance C faulty meter D no current E short circuit
-

Section B — Structured (15 marks)

Q6 (4 marks) — Series/parallel combination

A 24 V supply is connected to a $12\ \Omega$ resistor in series with two $24\ \Omega$ resistors in parallel.

- (a) Calculate the total resistance. (2)
- (b) Calculate the current in the $12\ \Omega$ resistor. (2)

Q7 (4 marks) — Thermistor modelling

A thermistor has resistance $8\ \text{k}\Omega$ at 20°C and $2\ \text{k}\Omega$ at 60°C .

- (a) Calculate current at 60°C when connected to 6 V. (2)
- (b) Explain why the resistance changes with temperature. (2)

Q8 (4 marks) — V-I experiment

A student plots a V-I graph for an unknown component and finds a curved line.

- (a) Suggest a component with this behaviour. (1)
- (b) Describe how to obtain an accurate graph. (3)

Q9 (3 marks) — Power & voltage

Calculate the voltage across a $50\ \Omega$ resistor when it dissipates 12.5 W.



CHALLENGE PAPER 2 — Answers

MCQs: 1 B, 2 B, 3 A, 4 C, 5 A

Q6: Parallel pair = $12\ \Omega$; total = $24\ \Omega$; $I = 24/24 = 1\ \text{A}$

Q7: $I = 6/2000 = 0.003\ \text{A}$; temperature increases $\rightarrow$ more charge carriers

Q8: Diode/Lamp; use multiple readings, controlled increments, stable supply

Q9: $V = \sqrt{PR} = \sqrt{12.5 \times 50} = 25 \text{ V}$

Audit: ✓ All standards met